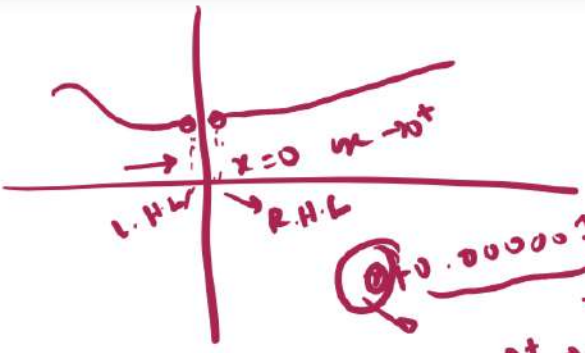

$0 + 0.000001$

$0^+ \to 0^+a$

$$\lim_{x\to 3} \frac{x^2-9}{x-3} = \frac{0}{0}$$

$$\frac{x-3}{x-3}$$

3^+

3^-

$$\lim_{x\to 0^-} f(x)$$

$$= \lim_{x\to 0^-} \frac{\tan kx}{x}$$

$$= \lim_{h\to 0} \frac{\tan k(0-h)}{0-h}$$

$x = 0 - h$

$x \to 0$

$h \to 0$

$h \to 0$

$kh \to 0$

$$= \lim_{kh\to 0} \frac{-\tan kh}{-hxk} \times k$$

$$= k$$

$2k^2 = k$

$k = 0, \frac{1}{2}$

$k = \frac{1}{2}$

$$\lim_{x \to 0} \frac{e^{x} - e^{-x} - 2x}{x - \sin x}$$

$$\frac{e^{0} - e^{-0} - 2 \times 0}{0 - \sin 0}$$

$0/0 \to$ L. Hôpital

$\frac{d}{dx} e^{-x} = -e^{-x}$

$$\lim_{x \to 0} \frac{e^{x} + e^{-x} - 2}{1 - \cos x}$$

$$\frac{e^{0} + e^{-0} - 2}{1 - 1} = 0/0$$

$\frac{d}{dx}(a^{x}) = (a^{x} \ln a)$

$$\lim_{x \to 0} \frac{e^{x} - e^{-x}}{+\sin x} \quad . \quad \frac{e^{0} - e^{-0}}{0} = 0/0$$

$$\lim_{x \to 0} \frac{e^{x} + e^{-x}}{\cos x} \qquad \frac{e^{0} + e^{-0}}{\cos 0} = 2$$

$$y = 1 - x + \frac{x^{2}}{2} - \frac{x^{3}}{3} + \cdots \infty$$

$$z = -y - \frac{y^{2}}{2} - \frac{y^{3}}{3} - \cdots \infty$$

$y = e^{-x}$

$\Rightarrow 1-e^{z} = e^{-x}$

$\Rightarrow \frac{1}{e^{x}} = 1-e^{z}$

$\Rightarrow e^{x} = \frac{1}{1-e^{z}}$

$\Rightarrow \ln e^{x} = \ln \frac{1}{1-e^{z}}$

$\Rightarrow x = \ln\left(\frac{1}{1-e^{z}}\right)$

$z = \ln(1-y)$

$\Rightarrow e^{z} = 1-y$

$\Rightarrow y = 1-e^{z}$

$\cos C - \cos D$

$= 2\sin\frac{C+D}{2}\,\sin\frac{D-C}{2}$

$$\lim_{x\to 0} \frac{\cos(\sin x)-\cos x}{x^4} \quad [0/0]$$

$$\Rightarrow \lim_{x\to 0} \frac{-\sin(\sin x)\cdot\cos x + \sin x}{3x^3} \quad [0/0]$$

$$\Rightarrow \lim_{x\to 0} -\cos($$

$$\Rightarrow \frac{2\sin\left(\frac{x+\sin x}{2}\right)\cdot\sin\left(\frac{x-\sin x}{2}\right)}{x^4}$$

65 70 80 85 → 145-146 90 out 34 20-25 c→33 32 1-2 1 300 230

$P \to 3a, -8, 3b$

$$\lim_{x\to 0} \frac{\sin(\text{masala})}{\text{masala}} = 1$$

$\text{masala} \to x, \sin x, \cos x$

$\sin(\text{masala}) \to x, \sin x, \cos x$

$$\Rightarrow \lim_{x\to 0} \frac{2\sin\left(\frac{x+\sin x}{2}\right)\sin\left(\frac{x-\sin x}{2}\right)}{x^4 \times \left(\frac{x+\sin x}{2}\right)\cdot\left(\frac{x-\sin x}{2}\right)} \times \left(\frac{x+\sin x}{2}\right) \times \left(\frac{x-\sin x}{2}\right)$$

$$2\lim_{x\to 0} \frac{\left(\frac{x^3-\sin^3 x}{4}\right)}{x^4} = \frac{1}{2}\lim_{x\to 0}\frac{x^3-\sin^3 x}{x^4}$$

$$= \frac{1}{2}\lim_{x\to 0}\frac{x^3\left(1-\frac{\sin^3 x}{x^3}\right)}{x^4}$$

$$\frac{1-\frac{\sin^3 x}{x^3}}{x^3}$$

$$\frac{1}{x^3} - \frac{\sin^3 x}{x^4}$$

$$\frac{1}{2}\times\frac{1}{3} = \frac{1}{6}$$

$$= \frac{1}{2}\lim_{x\to 0}\frac{1-\frac{\sin^3 x}{x^3}}{x^3}$$

$$\frac{x^2 - \sin^2 x}{x^4} \Rightarrow \frac{2x - 2\sin x \cos x}{4x^3}$$

$$\Rightarrow \frac{2x - \sin 2x}{4x^3}$$

$$\Rightarrow \frac{2 - 2\cos 2x}{12x^2}$$

$$\Rightarrow \frac{4\sin 2x}{24x}$$

$$\Rightarrow \frac{8\cos 2x}{24}$$

$$= \cancel{8}/\cancel{24}_3 = 1/3$$

$$\lim_{a \to 0} \frac{2}{\sqrt{2+\sqrt{2+2\cos 4a}}} \to a = 0$$

$$\frac{2}{\sqrt{2+\sqrt{4}}} \qquad 2/2 = 1$$

1^{∞} $0/$ undefined

$$\lim_{x \to \pi/2} (\sin x)^{\tan x} = y$$

$$\Rightarrow \ln y = \lim_{x \to \pi/2} \tan x \ln(\sin x)$$

$x - \pi/2 = h$

$x \to \pi/2$

$h \to 0$

$$\Rightarrow \ln y = \lim_{h \to 0} \tan(\pi/2 + h) \ln(\cos h)$$

$$= \lim_{h \to 0} \frac{h}{-\tan h} \times \frac{\ln \cos h}{h}$$

$$= \lim_{h \to 0} \frac{-\ln \cos h}{h} \qquad \frac{\ln 1 = 0}{h = 0} \left[\frac{0}{0}\right]$$

$1/x^2$

$d/dx\,(x^{-2})$

$-2(x^{-2-1})$

$-2x^{-3}$

$\frac{-2}{x^3}$

1^{∞}

$$= \lim_{h \to 0} -\frac{1}{\cos h}(-\sin h)$$

$$= \lim_{h \to 0} \tan h = 0 \qquad 1$$

$y = e^{0} = 1$

$\lim_{x \to \infty}$

$\frac{1}{e^{\infty}} = 0$ $\quad xe^{\infty}$ $\quad \infty$

$\lim_{n \to \infty} \frac{e^{x^2} \cdot 1}{ne^{x}}$

$\frac{e^{\infty}}{\infty}$ $\quad \infty/\infty$

$\frac{1}{e^{\infty}} = \frac{1}{\infty} = 0$ $\quad 0^{+}$

$\frac{2x\, e^{x^2}}{ne^{x}}$

$2x \cdot e^{x^2 - x}$

$2x\, e^{x(x-1)}$

$\infty \times \infty$

Undefined

$\cancel{\sqrt{x}}$

$\sqrt{x^2} \cdot \frac{1}{x}$

$x^2 \cdot \sqrt{\frac{1}{x}}$

$\frac{1}{\infty} = 0$ $\quad 0$

$$\frac{1 + \frac{x^2}{2} + \frac{\left(\frac{x^2}{2}\right)^2}{2!} - \cdots}{nx\left(1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots\right)}$$

$\lim_{x \to \infty} \left(\sqrt{x + \sqrt{x}} - \sqrt{x}\right)$

$\Rightarrow \frac{x + \sqrt{x} - x}{\sqrt{x + \sqrt{x}} + \sqrt{x}}$

$\sqrt{x}$

$$\frac{\cancel{\sqrt{x}}}{\cancel{\sqrt{x}}\sqrt{1+1/\sqrt{x}}+\cancel{\sqrt{x}}} \qquad \frac{\sqrt{x}}{\sqrt{x+\sqrt{x}}+\sqrt{x}}$$

$$= \lim_{x\to\infty} \frac{1}{1+\sqrt{1+1/\sqrt{x}}} \qquad \frac{1}{1+\sqrt{1+1/\sqrt{\infty}}} = \frac{1}{1+1} = \frac{1}{2}$$

$$-1 \le \sin x \le 1$$

$$\lim_{x\to\infty} \frac{3x^2-\sin 2x}{x^2+5} \quad \sin(\infty)x$$

$\sin\infty$

$\cos\infty$

masala $-1 \le$ Masala $-\sin 2x \le$ masala 1

$$\boxed{-1 \le \sin 2x \le 1}$$

$$-1\times -1 \ge -1\times \sin 2x \ge -1$$

$$1 \ge -\sin 2x \ge -1$$

$$\Rightarrow -1 \le -\sin 2x \le 1$$

$$\Rightarrow 3x^2-1 \le 3x^2-\sin 2x \le 3x^2+1$$

$$\frac{3x^2-1}{x^2+5} \le \frac{3x^2-\sin 2x}{x^2+5} \le \frac{3x^2+1}{x^2+5}$$

L.H.L $\lim_{x\to\infty} \frac{3x^2-1}{x^2+5}$

$\frac{3}{1} = 3$

R.H.L $\lim_{x\to\infty} \frac{3x^2+1}{x^2+5} = 3$

L.H.L = R.H.L

$$\lim_{x\to\infty} \frac{3x^2-\sin 2x}{x^2+5} = 3$$

x^2+5 x^2+5 x^2+5

L.H.L $\lim_{x\to\infty} \frac{3x^2-1}{x^2+5}$

$\frac{3}{1}=3$

R.H.L

L.H.L = R.H.L

$\lim_{x\to\infty} \frac{3x^2+1}{x^2+5} = 3$

$\lim_{x\to\infty} \frac{3x^2-\sin 2x}{x^2+5} = 3$

$\frac{x^2\left(3-\frac{\sin 2x}{x^2}\right)}{x^2\left(1+5/x^2\right)} \Rightarrow \frac{3-\frac{\sin\infty}{\infty^2}}{1+5/x^2}$ $\to 0$

$\frac{1}{\infty^2}$ $\frac{1}{\infty} = 0 \times \sin 2x = 0$

LDNE

$\frac{3}{1}$

$\sin 2\times\infty \to 0$

$\lim_{x\to\infty} \frac{3x^2-1}{x^2+5} \le \lim_{x\to\infty} \frac{3x^2-\sin 2x}{x^2+5}$

$\le \lim_{x\to\infty} \frac{3x^2+1}{x^2+5}$

$\lim_{x\to\infty} \frac{3x^2-\sin 2x}{x^2+5} = 3$

$3 \le \lim_{x\to\infty} \frac{3x^2-\sin 2x}{x^2+5} \le 3$

$\frac{e^{x^2}}{n\times e^x}$ $x\to\infty$

$\frac{1}{e^\infty} = 0 \times e^{x^2} = 0$